



Lithium-Ion Battery RUL Prediction Using CNN-LSTM Optimized by Improved Artificial Lemming Algorithm

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Accurate prediction of remaining useful life is paramount for effective lithium battery health management. This paper introduces a comprehensive prediction model that integrates an improved artificial lemming algorithm (ALA), a convolutional neural network (CNN), and a long short-term memory (LSTM) neural network. First, four sets of indirect health features were extracted from lithium battery charge-discharge cycle experiments to serve as predictive characteristics: constant current charge time, constant voltage rise time, and time integral of the constant voltage charge current curve during the charge phase, as well as constant voltage discharge time during the discharging phase. To mitigate the ALA's inherent susceptibility to local optima, an improved ALA algorithm was designed. This was achieved by incorporating the Tent chaotic map to enhance population diversity and integrating the mutation and crossover operations from the differential evolution algorithm, thereby improving both local search accuracy and convergence speed. Finally, a hybrid CNN-LSTM neural network model is proposed, capitalizing on the complementary strengths of both architectures. The model's predictive capability is further enhanced by employing optimization algorithms to address the sensitivity issues related to neural network hyperparameters.

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Lithium-ion batteries are increasingly applied in portable electronic devices, electric vehicles, and large-scale energy storage systems.¹ Lithium batteries are highly favored for their advantages including high power density, large energy density, low self-discharge rate, and long cycle life.² However, battery life and performance degradation—particularly the accurate prediction of remaining capacity—have emerged as critical industry challenges.³ Precise battery life prediction is essential for ensuring battery safety, performance, and reliability.⁴

Currently, research methods for lithium-ion battery life prediction primarily include model-driven and data-driven approaches.⁵ Model-driven methods primarily leverage physical, chemical, and thermodynamic principles to predict battery life. For example, Sadabadi K. K. et al. developed a remaining useful life (RUL) prediction algorithm based on parameter estimation of the enhanced single-particle model (eSPM).⁶ Wu R. et al. investigated an RUL prediction method utilizing variable modal decomposition and Gaussian process regression; however, this approach exhibits sensitivity to noise and limited adaptability.⁷ Furthermore, Luo R. et al.⁸ achieved rapid SOC and SOH prediction for retired batteries using electrochemical impedance spectroscopy (EIS), but their approach was only validated on retired lithium iron phosphate batteries with a rated capacity of 15.5 Ah, thus limiting its universality. In general, model-driven methods rely heavily on model accuracy, require significant development time, are susceptible to noise and environmental interference, and demonstrate poor generalization capabilities.⁹

In contrast, data-driven approaches depend on extensive experimental data to summarize patterns of battery performance changes, thereby predicting the remaining life. This method effectively overcomes the limitations of traditional model-based approaches, such as low accuracy, modeling difficulty, and poor generalizability.¹⁰ Current data-driven techniques include artificial neural networks,¹¹ support vector machines,¹² and deep learning.¹³ Recent advancements include Li R. et al.'s SSA-LSTM method for SOH prediction under variable modal decomposition,¹⁴ and Zhang Z. et al.'s novel SOH estimation integrating Kolmogorov-Arnold Networks (KAN) with Long Short-Term Memory (LSTM) networks.¹⁵ Zou G et al. proposed a hybrid lithium battery life prediction model based on convolutional neural networks (CNN) and long short-term memory networks (LSTM) to address the small-sample problem.¹⁶ Zeng F et al. addressed the limitations of single models in modeling capability and the weak feature coordination in hybrid models by proposing a hybrid deep learning model combining CNN, Transformer, and LSTM.¹⁷ Additionally, Wang et al. improved the Sparrow Optimization

Algorithm (SSA) with a logistic chaotic map and dynamic weight factors, using the resulting ISSA to optimize LSTM hyperparameters for significantly enhanced prediction accuracy.¹⁸

Additionally, optimization algorithms play a crucial role in hyperparameter tuning for models. Tao Q et al. proposed a predictive approach combining the Whale Optimization Algorithm (WOA) with a Gated Recurrent Unit (GRU) featuring an attention mechanism.¹⁹ Chen K et al. employed Extreme Learning Machines (ELM) to establish a State of Health (SOH) estimation model for lithium batteries, utilizing the Particle Swarm Optimization (PSO) algorithm to optimize the model's hyperparameters.²⁰ The Artificial Lemming Algorithm (ALA) is a novel bio-inspired intelligent optimization algorithm proposed in 2025.²¹ Compared to traditional algorithms such as the particle swarm optimization,²² whale optimization algorithm,²³ and genetic algorithm,²⁴ the ALA optimization algorithm demonstrates superior performance optimization and convergence speed. However, it also carries the risk of getting stuck in local optima.

To address the aforementioned issues, this paper proposes an improved Artificial Lemming Algorithm (ALA) combined with a hybrid model integrating Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks. This approach tackles the capacity regeneration phenomenon and hyperparameter sensitivity in battery life prediction. By incorporating the Tent chaotic mapping into the original algorithm to enhance population diversity and initial solution randomness, and further introducing mutation and crossover operations from the Differential Evolution (DE) algorithm at each iteration to prevent getting stuck in local optima, combined with the advantages of CNN and LSTM, an RUL prediction model based on the improved ALA-optimized CNN-LSTM model is established.

Battery Data Processing and Health Factor Extraction

Battery dataset.—This paper utilizes two major public datasets for lithium-ion battery research: the dataset provided by the National Aeronautics and Space Administration (NASA) and the charge-discharge cycle aging dataset from the Center for Advanced Life Cycle Engineering (CALCE) at the University of Maryland.

The NASA dataset features 18650 lithium-ion cells, each with a nominal capacity of 2 Ah. The four specific batteries used for testing are B05, B06, B07, and B18.

The CALCE dataset includes two lithium cells, CS35 and CS36, each having a nominal capacity of 1.1 Ah.

The specific testing process is as follows:

NASA's specific testing procedure: (1) Charging process: At room temperature (24 °C), perform constant-current charging at 1.5 A until the battery voltage reaches 4.2 V. Then switch to

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constant-voltage mode and continue charging until the charging current drops below 20 mA, at which point charging is terminated. (2) Discharge Process: All lithium-ion batteries undergo constant-current discharge at 2 A until voltages at terminals B05, B06, B07, and B18 drop to 2.7 V, 2.5 V, 2.2 V, and 2.5 V respectively. (3) When the battery undergoes multiple repeated charge-discharge cycles and reaches end-of-life (EOL), the test is terminated. The battery reaches EOL at 70% of its rated capacity (1.4 Ah). Since B07 battery does not drop to the lower limit, its EOL condition is set at 72% of rated capacity (1.44 Ah).

CALCE Specific Testing Procedure: (1) Charging Process: At 24 °C room temperature, perform constant-current charging at a rate of 0.5 C until the battery voltage reaches 4.2 V. Subsequently, switch to constant-voltage mode and continue charging until the charging current drops below 50 mA, then terminate charging. (2) Discharging Process: All lithium-ion batteries undergo constant-current discharging at a rate of 1 C until the voltage drops below 2.7 V. (3) The test concludes when the battery reaches end-of-life (EOL) after multiple charge-discharge cycles. In this study, EOL was defined as when the battery capacity reached 70% of its rated capacity, corresponding to a capacity threshold of 0.8 Ah. Figure 1 illustrates the capacity degradation curves of NASA and CALCE batteries vs cycle count.

Battery health factor extraction.—This paper extracted four sets of HF from the lithium battery dataset. Three HF were selected for the charging phase: constant current charging time (CCCT), constant voltage rise time (3.9 V~4.1 V), and the time integral of the current curve during the constant voltage charging stage. Characteristic representation during the discharging phase is also essential; thus, the constant voltage decay time (4 V~3.9 V) during the discharging phase was extracted as the fourth HF. These are designated as HF1, HF2, HF3, and HF4, as shown in Fig. 2 below:

Relevance analysis.—This paper employs the Pearson Correlation Coefficient (PCC) to analyze the correlation between the extracted health factor HF and battery capacity, calculated as follows:

$$Person = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad [1]$$

In the formula, Person's correlation coefficient ranges from -1 to 1 ; n represents the sample size; x_i and y_i denote the i th data point; $\bar{x}$ and $\bar{y}$ denote the mean values of the variables. The closer the absolute value of the correlation coefficient approaches 1, the stronger the correlation.

As shown in Table I, among the selected HF components in this study, HF1, HF2, and HF4 exhibit positive correlations, while HF3 shows a negative correlation. The absolute values of the correlation coefficients all exceed 0.85, indicating strong correlations.

Table I. HF and capacity correlation coefficients.

Battery	HF1	HF2	HF3	HF4
B05	0.9989	0.9956	-0.9915	0.9707
B06	0.9969	0.9933	-0.9667	0.9642
B07	0.9988	0.9894	-0.9698	0.9783
B18	0.9948	0.9891	-0.8720	0.9987
CS35	0.9990	0.9621	-0.9156	0.9099
CS36	0.9993	0.9528	-0.9136	0.9276

RUL Prediction Method Based on an Improved Artificial Lemming Optimization CNN-LSTM Approach

Convolutional neural network.—Convolutional neural networks (CNN) first extract local features from input data efficiently through convolutional layers, then refine these features further via pooling layers, and finally map them to the final output space through fully connected layers,²⁵ as shown in Fig. 3.

Long-term and short-term memory neural network.—By introducing a gating mechanism, LSTM overcomes the gradient explosion and vanishing gradient issues inherent in traditional recurrent neural networks, enabling it to better capture long-term dependencies in time series data. This characteristic makes it highly suitable for tasks involving long-term dependencies.²⁶ The LSTM system architecture is shown in Fig. 4.

- (1) Forgetting Gate: Determines which information in the current hidden state should be discarded. Using a sigmoid activation function, it outputs values between 0 and 1 representing the importance of each memory cell. A value of 0 indicates complete discard, while a value of 1 indicates complete retention.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad [2]$$

In the formula, f_t is the forget gate, σ is the sigmoid function, W_f and b_f are the weight matrix and bias matrix of the forget gate, respectively. h_{t-1} is the unit state output at time step $t-1$, x_t is the unit state input at time step.

- (2) Input Gate: Controls which new information can be added to the memory cell. This process occurs in two steps: first, the sigmoid activation function determines which information to update; second, the tanh activation function generates new candidate values to determine the content added to the memory cell.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad [3]$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad [4]$$

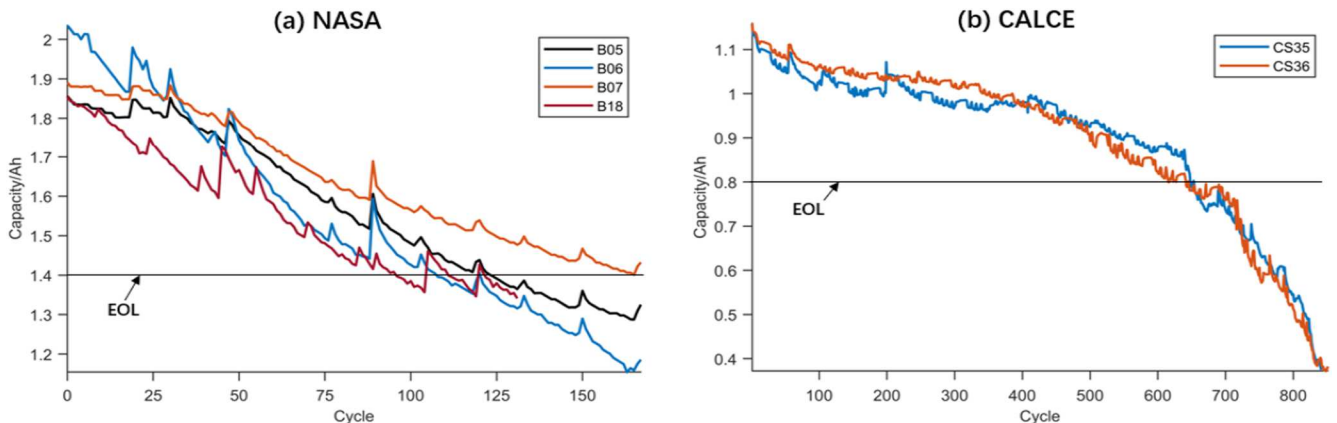


Figure 1. (a) NASA battery capacity decay curve (b) CALCE battery capacity decay curve.

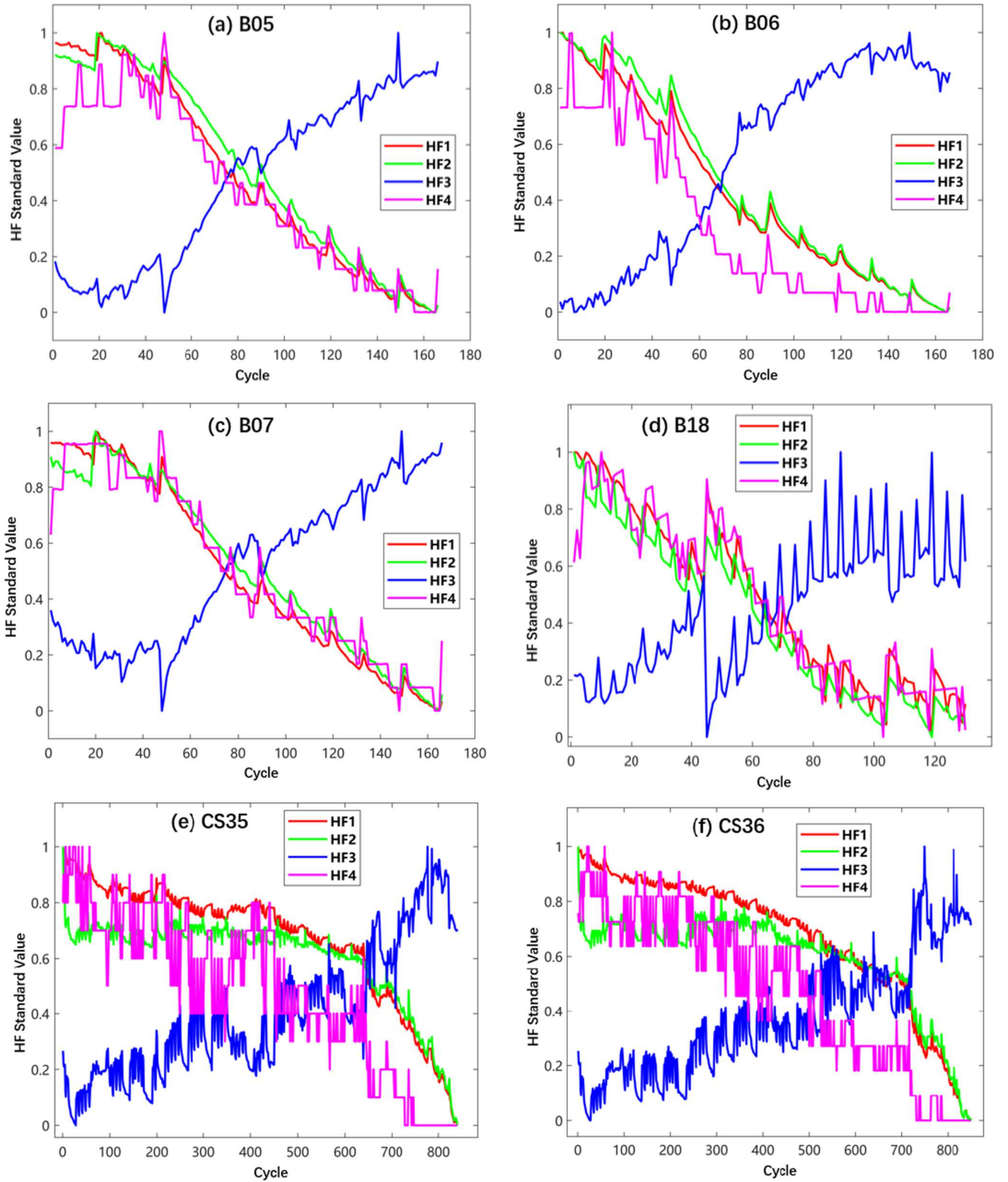


Figure 2. Health factors extracted by NASA and CALCE.

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad [5]$$

In the formula, i_t is the input gate, W_i and b_i are the weight matrix and bias matrix of the input gate, respectively. $\tilde{C}_t$ is the candidate memory cell, C_t and C_{t-1} represent the cell states at time steps t and $t-1$, respectively.

(3) Output Gate: Based on the current input information, it determines the output value for the next time step. The output gate filters the information to be output through the sigmoid activation function, then adjusts it via the tanh activation function to keep the output value within an appropriate range.

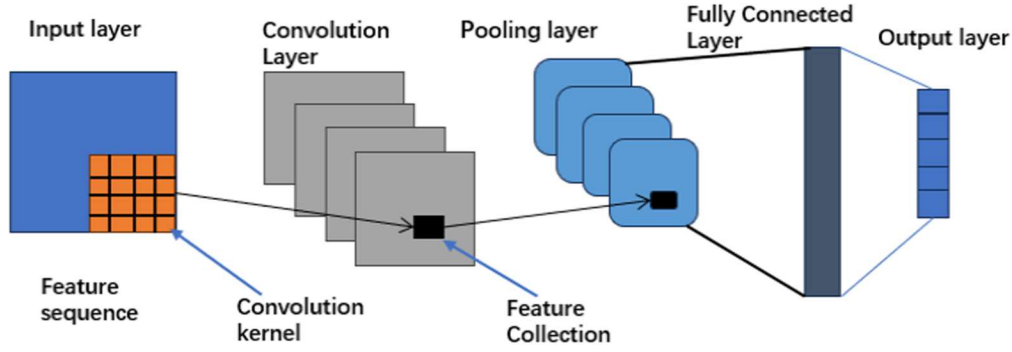


Figure 3. Basic structure diagram of a CNN.

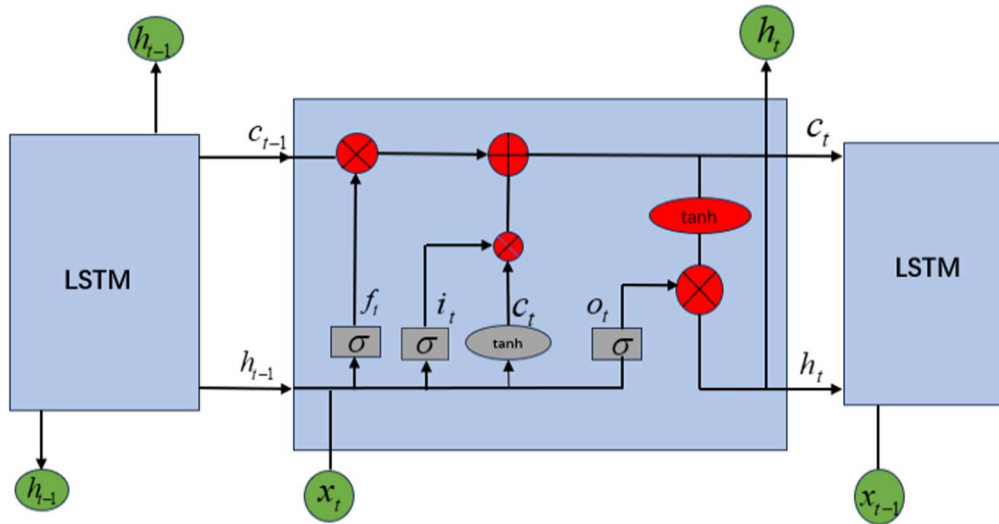


Figure 4. LSTM neural network architecture diagram.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad [6]$$

$$h_t = o_t * \tanh(C_t) \quad [7]$$

In the formula, o_t is the output gate, W_o and b_o are the weight matrix and bias matrix of the output gate, h_t is the output of the LSTM at time step.

Improved artificial lemming optimization algorithm.—Standard artificial hamster optimization algorithm.—

(1) Initialize the population

ALA is an intelligent optimization algorithm based on swarm search. Before entering the iterative process, it requires initializing the positions of all lemmings. The entire initial candidate solution set is represented in matrix form:

$$\vec{Z} = \begin{bmatrix} z_{1,1} & z_{1,2} & \cdots & z_{1,Dim} \\ z_{2,1} & z_{2,2} & \cdots & z_{2,Dim} \\ \vdots & \vdots & \vdots & \vdots \\ z_{N,1} & z_{N,2} & \cdots & z_{N,Dim} \end{bmatrix} \quad [8]$$

The initial population is represented by the following formula:

$$z_{i,j} = LB_j + rand \times (UB_j - LB_j) \quad [9]$$

$$i = 1, 2, \dots, N; j = 1, 2, \dots, Dim$$

In the formula: $rand$ is a random number within the range of 0 to 1; UB_j is the upper bound for solving the problem in the j dimension,

and LB_j is the lower bound for solving the problem in the j dimension.

(2) Long-distance migration behavior

In natural environments, when food resources become scarce, lemmings undertake random long-distance migrations in search of more favorable survival conditions and resources. In the ALA algorithm, these long-distance migrations correspond to the global exploration phase. By simulating lemming migration behavior, the algorithm conducts extensive searches across the solution space to discover potentially high-quality regions. To model this behavior, the following equations are proposed:

$$\vec{Z}_i(t+1) = \vec{Z}_{best}(t) + F \times \vec{BM} \times \begin{pmatrix} \vec{R} \times (\vec{Z}_{best}(t) - \vec{Z}_i(t)) \\ +(1 - \vec{R}) \times (\vec{Z}_i(t) - \vec{Z}_a(t)) \end{pmatrix} \quad [10]$$

In the formula: $\vec{Z}_i(t+1)$ denotes the position of the i th lemming at iteration $t+1$, $\vec{Z}_i(t)$ denotes the position of the i th lemming in the t th iteration, $\vec{Z}_{best}(t)$ indicates the optimal solution at the current position, $\vec{Z}_a(t)$ represents the position of a randomly selected search individual within the lemming population. a is an integer index between 1 and N , where F is used to alter the direction of exploration, $\vec{BM}$ denotes Brownian motion, which explores potential regions in space by utilizing dynamic and uniform strides, $\vec{R}$ is a random vector of size $1 \times Dim$, whose elements are random numbers uniformly distributed over the interval $[-1, 1]$. It is specifically represented as follows:

$$F = \begin{cases} 1 & \text{if } [2 \times rand + 1] = 1 \\ -1 & \text{if } [2 \times rand + 1] = 2 \end{cases}$$

$$f_{BM}(x; 0, 1) = \frac{1}{\sqrt{2\pi}} \times \exp\left(-\frac{x^2}{2}\right)$$

$$\vec{R} = 2 \times rand(1, Dim) - 1 \quad [11]$$

(3) Undermining behavior

After locating a suitable habitat, lemmings excavate burrows as shelters, with this digging behavior corresponding to the local development phase. Lemmings randomly select digging locations based on the current burrow's position and the positions of randomly selected individuals within the population. This strategy not only provides secure dwellings but also helps lemmings evade predator threats. The position update formula is as follows:

$$\vec{Z}_i(t+1) = \vec{Z}_i(t) + F \times L \times (\vec{Z}_{best}(t) - \vec{Z}_b(t)) \quad [12]$$

In the formula: $\vec{Z}_b$ denotes a search individual randomly selected from the lemming population, b is a random integer index between 1 and N . The calculation formula for L is as follows:

$$L = rand \times \left(1 + \sin\left(\frac{t}{2}\right)\right) \quad [13]$$

(4) Foraging behavior

Lemmings forage near their habitats to secure sufficient food resources. Foraging behavior corresponds to searching within a local area to further optimize solutions. The spiral winding mechanism simulates the foraging behavior of lemmings in their habitat burrows, with the position update formula as follows:

$$\vec{Z}_i(t+1) = \vec{Z}_{best}(t) + F \times spiral \times rand \times \vec{Z}_i(t) \quad [14]$$

In the formula: *spiral* denotes the spiral shape of random search during foraging, *radius* represents the radius of the foraging range, i.e., the Euclidean distance between the current position and the optimal solution. It is composed of the following formula:

$$spiral = radius \times (\sin(2 \times \pi \times rand) + \cos(2 \times \pi \times rand)) \quad [15]$$

$$radius = \sqrt{\sum_{j=1}^{Dim} (z_{best,j}(t) - z_{i,j}(t))^2} \quad [16]$$

(5) Avoidance of Predators

When faced with danger, lemmings flee swiftly, making burrows their natural refuge. This predator-avoidance behavior corresponds to an algorithm's mechanism for escaping a local optimum during search—by jumping out of the current region when encountering suboptimal solutions or becoming trapped in a local optimum, thereby preventing premature convergence.

$$\vec{Z}_i(t+1) = \vec{Z}_{best}(t) + F \times G \times Levy(Dim) \times (\vec{Z}_{best}(t) - \vec{Z}_i(t)) \quad [17]$$

$$G = 2 \times \left(1 - \frac{t}{T_{max}}\right) \quad [18]$$

In the formula: G is the lemming escape coefficient, which gradually decreases with increasing iteration count; *Levy* is the Lévy walk function, simulating deceptive maneuvers during lemming

escape; T_{max} denotes the maximum iteration count. The stride formula for the Lévy walk is as follows:

$$Levy(x) = 0.01 \times \frac{u \times \sigma}{|v|^{\frac{1}{\beta}}}$$

$$\sigma = \left(\frac{\Gamma(1+\beta) \times \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times 2^{\left(\frac{\beta-1}{2}\right)}} \right)^{\frac{1}{\beta}} \quad [19]$$

In the formula: u and v are random values within the interval $[0,1]$, β is a constant with a value of 1.5.

Improved artificial lemming optimization algorithm.—This paper primarily addresses two aspects: introducing the Tent chaotic map during the algorithm's initialization phase to enhance population diversity and the randomness of solution initialization, while incorporating DE's mutation and crossover operations in each generation iteration to improve the algorithm's local search accuracy and convergence speed.

(1) Tent Chaos Mapping

The original ALA optimization algorithm employs conventional random initialization, which may result in an uneven distribution of the initial population and increase the likelihood of the algorithm getting stuck in local optima. By utilizing the Tent chaotic map to generate the initial population, the solution is uniformly distributed across the search space due to the chaotic map sequence's properties of ergodicity and randomness. This reduces the probability of getting stuck in local optima and shortens the initial optimization time. The formula for the Tent chaotic map is as follows:

$$x_{n+1} = \begin{cases} r \cdot x_n & \text{if } x_n < 0.5 \\ r \cdot (1 - x_n) & \text{if } x_n \geq 0.5 \end{cases} \quad [20]$$

In the formula: $x_n \in [0,1]$ is the n th value of the chaotic sequence, r is the control parameter. Map the values of the Tent chaos sequence to the variable range of the optimization problem $[lb, ub]$:

$$X_{(i,j)} = x_n(ub_j - lb_j) + lb_j \quad [21]$$

In the formula: $X_{(i,j)}$ is the j th dimensional variable of the i th individual, ub_j and lb_j represent the upper and lower bounds of the j th dimension, respectively.

(2) Differential Evolution Algorithm

Prior to updating the original ALA optimization algorithm, incorporating the core operations of mutation and crossover from DE introduced DE fusion. This effectively performs “survival of the fittest” and “genetic recombination” on the population during each iteration, significantly enhancing the algorithm's ability to escape local optima. By leveraging information from multiple individuals within the population to generate new solutions, it strengthens the algorithm's information exchange capabilities and improves local search precision. The crossover probability CR of the DE strategy is set to a fixed value of 0.7, and the scaling factor F is configured as a fixed value of 0.5. The determination of F and CR is critical to the reliability of the DE algorithm, and fixed values are suitable for rapid implementation.

1. Mutation Operations

Three distinct individuals are randomly selected from the population to undergo mutation, generating a mutated individual. Subsequently, crossover operations are performed to produce trial individuals. These trial individuals are compared with the current individual based on fitness, and their positions are updated accordingly. As per the following formula:

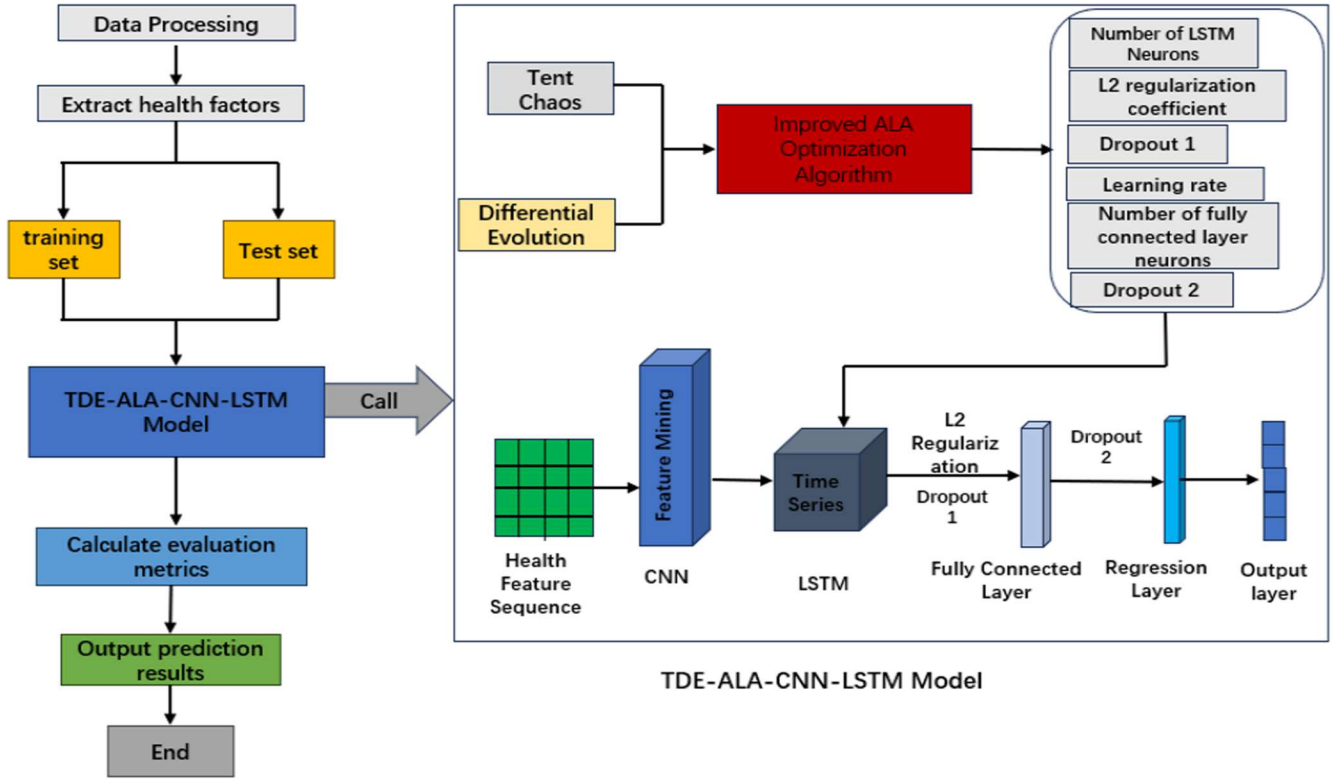


Figure 5. TDE-ALA-CNN-LSTM model architecture diagram.

$$v_{ij}(t) = x_{aj}(t) + F(x_{bj}(t) - x_{cj}(t)), j = 1, 2, \dots, D \quad [22]$$

Among them, $x_a(t) = (x_{a1}(t), x_{a2}(t), \dots, x_{aD}(t))$, $x_b(t) = (x_{b1}(t), x_{b2}(t), \dots, x_{bD}(t))$, $x_c(t) = (x_{c1}(t), x_{c2}(t), \dots, x_{cD}(t))$, are three randomly selected distinct individuals from the population where $a \neq b \neq c \neq i$, $x_{aj}(t)$, $x_{bj}(t)$ and $x_{cj}(t)$ represent the j th component of individuals, a , b and c , respectively; F is the mutation factor, typically taking values within the range $[0, 2]$, $v_i(t)$ denotes the resulting mutant individual.

2. Cross-operation

Cross the mutant individual $v_i(t)$ with the current individual $x_i(t)$ to generate the test individual $u_i(t) = (u_{i1}(t), u_{i2}(t), \dots, u_{iD}(t))$.

$$u_{ij}(t) = \begin{cases} v_{ij}(t), & \text{rand}(j) \leq CR \text{ or } j = \text{randn}(i) \\ x_{ij}(t), & \text{rand}(j) > CR \text{ and } j \neq \text{randn}(i) \end{cases} \quad [23]$$

3. Select Operation

Compare the fitness of individual $u_i(t)$ with that of current individual $x_i(t)$, retain the superior solution, and generate the next generation individual $x_i(t+1)$ according to the following formula:

$$x_i(t+1) = \begin{cases} u_i(t), & f(u_i(t)) < f(x_i(t)) \\ x_i(t), & f(u_i(t)) \geq f(x_i(t)) \end{cases} \quad [24]$$

Here, f denotes the fitness function.

Based on the TDE-ALA-CNN-LSTM model.—This paper proposes a hybrid CNN-LSTM neural network model that fully leverages the strengths of both CNN and LSTM neural networks in processing different types of data. CNNs excel at extracting spatial features from raw input data, while LSTMs are well-suited for processing time-series

data, enabling them to capture long-term dependencies within the data. By connecting CNNs and LSTMs in series, the CNN model first performs spatial feature extraction on the input data, uncovering deeper relationships within the data. The LSTM then receives these extracted spatial features as input, further learning the temporal relationships among these features to enhance the model's predictive performance.

Key steps: First, within the CNN module, create a one-dimensional convolutional layer. The core function of this layer is to perform local feature extraction on the input battery feature data. The size of the convolution kernel is $[1,1]$, The quantity is 2, Step size is $[1,1]$, Batch normalization is applied to the convolutional layers, and the ReLU activation function is introduced to enhance the model's expressive power. After passing through the ReLU activation, the features enter the pooling layer to achieve dimensionality reduction while strengthening the robustness of key features and filtering redundant information; The data is then flattened into a one-dimensional sequence through sequence unfolding and flattening layers, serving as input for the LSTM. The LSTM handles long-term dependencies in time series data. A sufficient number of hidden neurons is set to ensure the model's ability to learn sequence information. L2 regularization is applied to constrain parameter scale by penalizing weights, preventing overfitting in the LSTM layer. A Dropout1 layer is then added with a random dropout rate to reduce the model's tendency to overfit local features, thereby enhancing its generalization capability. Finally, a fully connected layer integrates the temporal features output by the LSTM to further uncover global correlations among features. To further suppress overfitting, a Dropout2 layer is added after the fully connected layer, followed by a regression layer that ultimately outputs the prediction results.

The TDE-ALA optimization algorithm employed in this paper optimizes the hyperparameters of the CNN-LSTM model. Six sets of hyperparameters requiring optimization were defined, including: - Number of hidden neurons in the LSTM layer - Initial learning rate - Dropout 1 random dropout rate after the LSTM layer - L2 regularization coefficient - Number of neurons in the fully connected layer - Dropout 2 random dropout rate after the fully connected layer The flowchart is shown in Fig. 5:

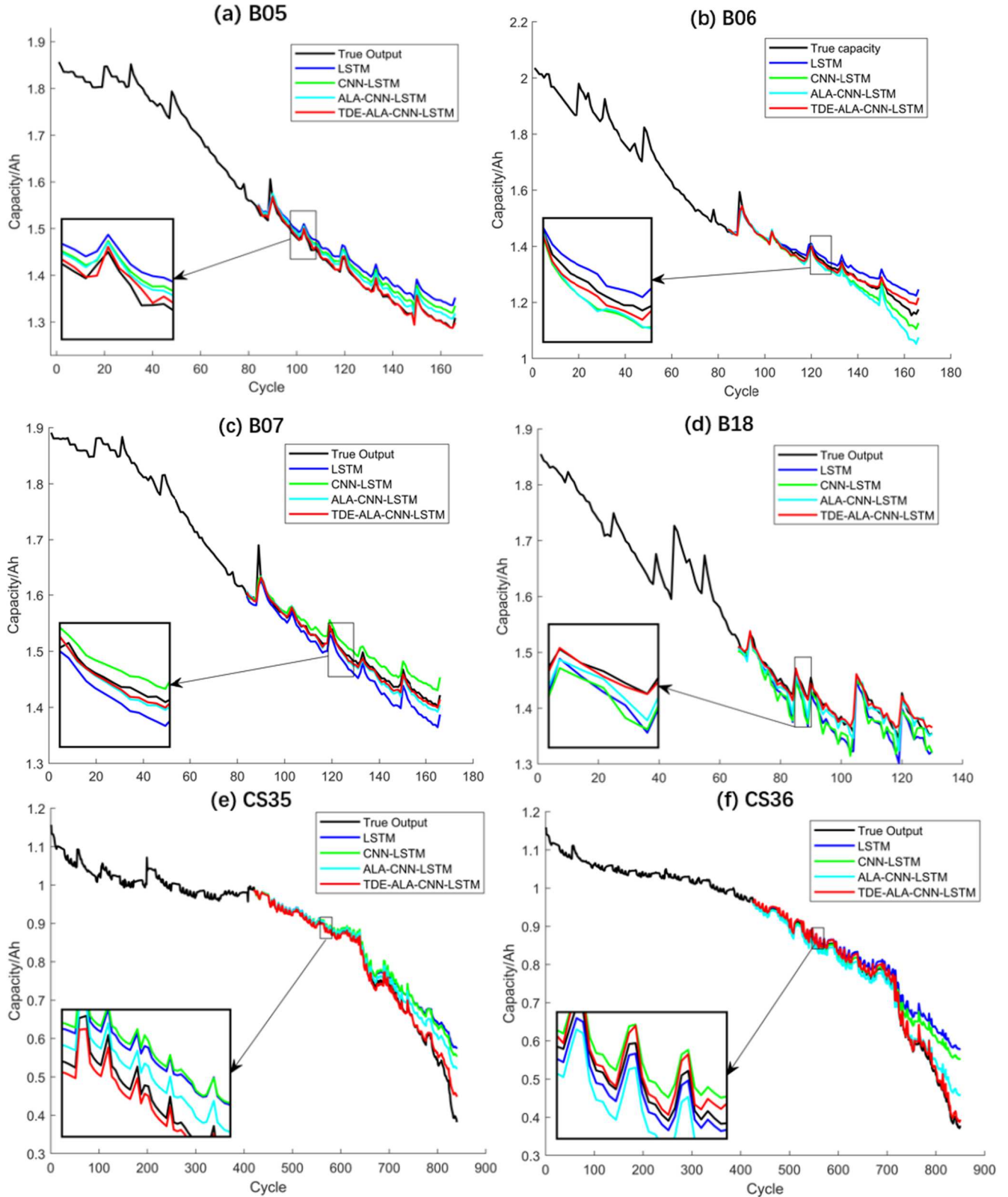


Figure 6. Prediction results for four different methods from NASA and CALCE batteries.

Experimental Validation and Comparative Analysis

Evaluation indicators.—To evaluate the performance of the proposed method, the Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and coefficient of determination were selected R^2 . The closer the RMSE and MAE values are to zero,

the higher the accuracy of the proposed method.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad [25]$$

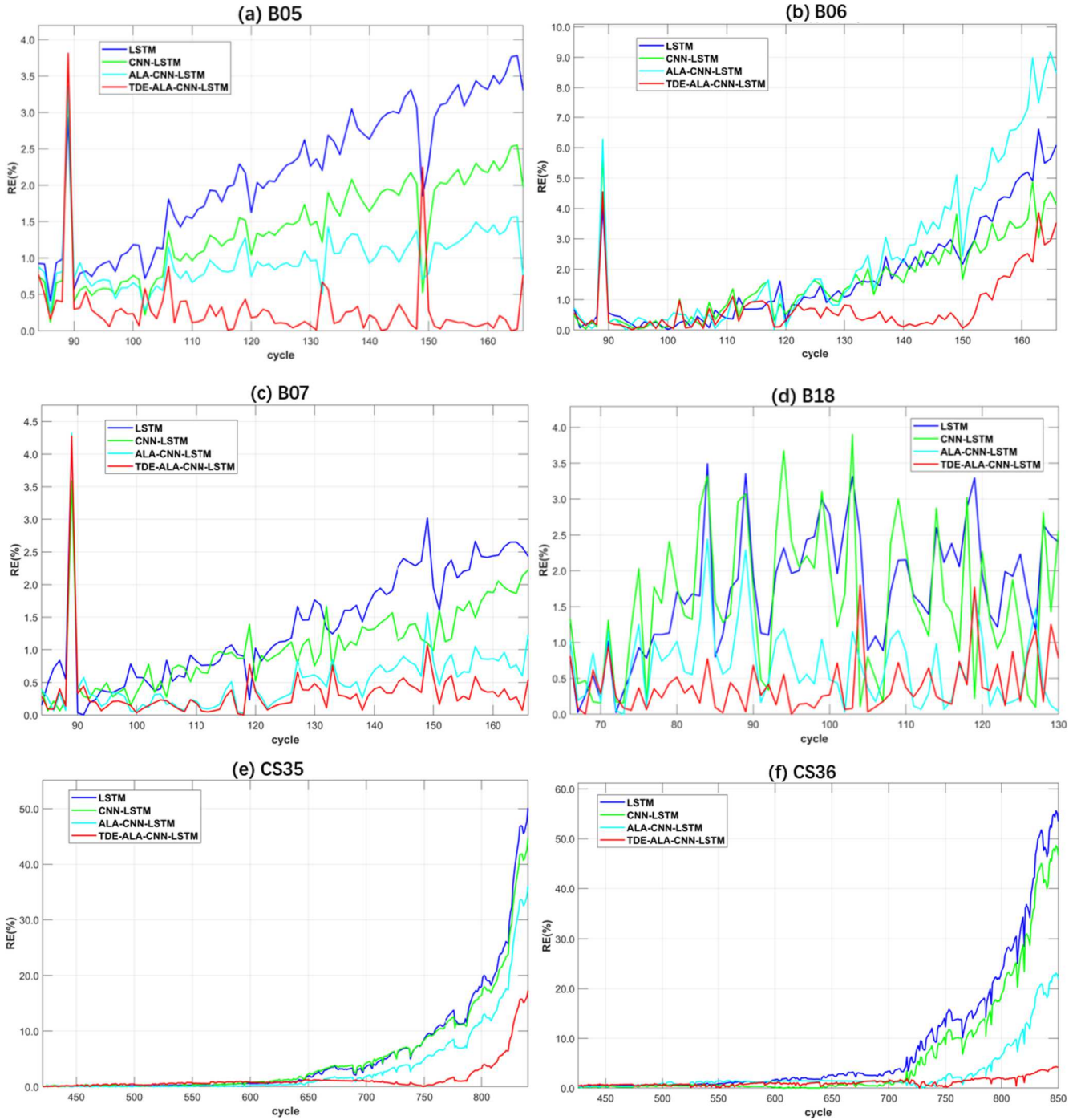


Figure 7. Relative Error Results for Four Different Methods from NASA and CALCE Batteries.

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad [26]$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad [27]$$

Here, n represents the number of samples, i denotes the i th sample, $\hat{y}_i$ and y_i denote the predicted value and true value of the i th sample, respectively.

Experimental analysis.—To validate the model's accuracy, the first 50% of the lithium battery aging lifecycle was selected as the training set, while the latter 50% served as the test set. For different batteries, four models—LSTM (blue curve), CNN-LSTM (green curve), ALA-CNN-LSTM (cyan curve), and TDE-ALA-CNN-LSTM (red curve)—were employed for longitudinal comparison to evaluate the predictive precision of the proposed model. Figure 6 below compares the prediction results of the four different methods, while Fig. 7 shows the relative error comparison within the prediction period:

Analysis of Relative Error Results: In batteries B05, B06, and B07, the early-stage relative error curves exhibit sudden spikes followed by rapid declines. This occurs because lithium batteries

Table II. Prediction metrics for four different methods of NASA and CALCE batteries.

Battery	Method	RMSE	MAE	R^2
B05	LSTM	0.0318	0.0297	0.8432
	CNN-LSTM	0.0213	0.0195	0.9299
	ALA-CNN-LSTM	0.0148	0.0137	0.9659
	TDE-ALA-CNN-LSTM	0.0086	0.0042	0.9894
B06	LSTM	0.0298	0.0222	0.9079
	CNN-LSTM	0.0263	0.0203	0.9286
	ALA-CNN-LSTM	0.0421	0.0297	0.8166
	TDE-ALA-CNN-LSTM	0.0152	0.0098	0.9761
B07	LSTM	0.0238	0.0202	0.8645
	CNN-LSTM	0.0176	0.0152	0.9256
	ALA-CNN-LSTM	0.0115	0.0078	0.9683
	TDE-ALA-CNN-LSTM	0.0094	0.0051	0.9787
B18	LSTM	0.0266	0.0240	0.6948
	CNN-LSTM	0.0263	0.0224	0.7009
	ALA-CNN-LSTM	0.0121	0.0096	0.9370
	TDE-ALA-CNN-LSTM	0.0078	0.0057	0.9741
CS35	LSTM	0.0532	0.0311	0.8856
	CNN-LSTM	0.0492	0.0301	0.9023
	ALA-CNN-LSTM	0.0352	0.0190	0.9498
	TDE-ALA-CNN-LSTM	0.0140	0.0087	0.9921
CS36	LSTM	0.0697	0.0434	0.8213
	CNN-LSTM	0.0567	0.0306	0.8818
	ALA-CNN-LSTM	0.0243	0.0160	0.9783
	TDE-ALA-CNN-LSTM	0.0081	0.0076	0.9976

experience capacity regeneration during specific cycles, causing abrupt changes in health characteristics. This phenomenon affects all models, leading to significant fluctuations in prediction errors. However, it does not invalidate the overall conclusions. Subsequently, the batteries enter the capacity degradation phase where internal side reactions intensify, causing non-linear changes in health characteristics. This increases the difficulty of model fitting and gradually widens prediction errors. However, the proposed TDE-ALA-CNN-LSTM model, while also experiencing prediction error fluctuations due to these factors, maintains relatively small errors during the later prediction stages for Batteries B05 and B07. For Battery B06, the relative error increase remains relatively gradual. This demonstrates the model's superior disturbance resistance compared to other models. Battery B18 experienced multiple cycles with sudden real capacity changes due to capacity regeneration during prediction, complicating accurate forecasting. Yet, the relative error comparison chart shows the TDE-ALA-CNN-LSTM model exhibits smaller fluctuation amplitude compared to the other three models, indicating more stable predictive capability. For batteries CS35 and CS36, the curves reveal that prediction errors approached zero during the initial cycling phase. As cycle counts increased, prediction errors exhibited a steady upward trend. In the final stage before battery failure, errors increased rapidly. This aligns with the nonlinear characteristics of lithium battery capacity degradation and the evolving difficulty of RUL prediction. This demonstrates that the improved artificial lemming algorithm for CNN-LSTM hyperparameter optimization effectively enhances the model's prediction accuracy and stability. Particularly during the strongly nonlinear decay phase in the battery's final stage, the model maintains low prediction errors, exhibiting superior robustness.

Maximum Error Analysis: For batteries B05, B06, and B07, the maximum prediction error occurred during the early prediction phase. This directly correlates with the capacity regeneration phenomenon observed during the initial cycling period, where sudden capacity fluctuations caused temporary deviations between model predictions and actual values. For battery B18, the maximum prediction error occurred during the mid-prediction phase, stemming from multiple capacity regeneration events during this period. The

repeated fluctuations in actual capacity significantly increased prediction difficulty. For batteries CS35 and CS36, the maximum prediction error occurred during the late prediction phase. At this stage, intense internal side reactions caused health characteristics to exhibit strong nonlinear degradation, representing the core challenge phase for RUL prediction.

As shown in the table above, the RMSE and MAE values of the prediction results from the TDE-ALA-CNN-LSTM model proposed in this paper outperform those of the other three models, and the coefficient of determination also demonstrates a higher level. The four magnified sections in Fig. 6 reveal that LSTM and CNN-LSTM primarily focus on capturing the trend of battery capacity degradation, exhibiting weaker ability to capture local features. In contrast, the proposed TDE-ALA-CNN-LSTM model not only effectively fits the capacity degradation trend but also precisely captures local features, significantly enhancing prediction accuracy. Table II shows that the root mean square error (RMSE) of the ALA-CNN-LSTM model for B06 battery is significantly higher than that of the CNN-LSTM model, indicating that the optimization algorithm is prone to local optima. In contrast, the TDE-ALA-CNN-LSTM model exhibits the lowest error, demonstrating that the proposed improvement combining Tent chaos mapping with DE's differential evolution operation effectively avoids the issue of the ALA optimization algorithm getting stuck in local optima.

Experimental Conclusions

This paper proposes a novel lithium-ion battery life prediction model based on TDE-ALA-CNN-LSTM. Battery charge-discharge cycle aging datasets from NASA and CALCE were selected and processed, extracting four sets of HF as inputs for the prediction model, with battery capacity as the output. First, PCC analysis was employed to examine the correlation between health factors and battery capacity. Subsequently, leveraging the strengths of both CNN and LSTM models, a combined CNN-LSTM model was constructed. CNN effectively extracts local features from input data, while LSTM specializes in processing time-series data. Their combination simultaneously captures local features and global trends. Considering the significant impact of model hyperparameters on prediction outcomes, this paper also proposes an improved ALA optimization algorithm to address hyperparameter sensitivity issues. Building upon the original ALA algorithm, this approach incorporates the Tent chaotic map to enhance population diversity and initial solution randomness. It further integrates DE's mutation and crossover operations to prevent convergence to local optima. Ultimately, a prediction model based on TDE-ALA-CNN-LSTM was constructed.

Experimental results demonstrate that through longitudinal model comparisons of NASA and CALCE battery life predictions (using LSTM, CNN-LSTM, ALA-CNN-LSTM, and TDE-ALA-CNN-LSTM models respectively), the proposed model outperforms other models in prediction precision and accuracy, validating its effectiveness and precision. In the predictions for B06, B18, CS35, and CS36 batteries, the original optimization algorithm exhibited lower prediction accuracy, whereas the improved optimization algorithm significantly enhanced prediction accuracy. This further demonstrates that the improved optimization algorithm effectively avoids the issue of local optima.

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